

Aim: Transfer Learning using VGG16

```
from google.colab import files
import zipfile
import tensorflow as tf

uploaded=files.upload()

zip_file=list(uploaded.keys())[0]

with zipfile.ZipFile(zip_file,"r") as zip_ref:
    zip_ref.extractall("/content")

dataset_path="/content/dataset"

train_data=tf.keras.preprocessing.image_dataset_from_directory(
    dataset_path,
    validation_split=0.2,
    subset="training",
    seed=42,
    image_size=(224,224),
    batch_size=8
)

test_data=tf.keras.preprocessing.image_dataset_from_directory(
    dataset_path,
    validation_split=0.2,
    subset="validation",
    seed=42,
    image_size=(224,224),
    batch_size=8
)

num_classes=len(train_data.class_names)

base_model=tf.keras.applications.VGG16(
    weights="imagenet",
    include_top=False,
    input_shape=(224,224,3)
)
```

```

base_model.trainable=False

model=tf.keras.Sequential([
    tf.keras.layers.Rescaling(1./255),
    base_model,
    tf.keras.layers.GlobalAveragePooling2D(),
    tf.keras.layers.Dense(32,activation="relu"),
    tf.keras.layers.Dense(num_classes,activation="softmax")
])

model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

model.summary()

model.fit(
    train_data,
    validation_data=test_data,
    epochs=5
)

loss,accuracy=model.evaluate(test_data)

print("Accuracy:",accuracy)

```

Output:

small_cnn_dataset.zip(application/x-zip-compressed) - 19105 bytes, last modified: 7/17/2026 - 100% done

Saving small_cnn_dataset.zip to small_cnn_dataset.zip

Found 40 files belonging to 2 classes.

Using 32 files for training.

Found 40 files belonging to 2 classes.

Using 8 files for validation.

Downloading data from

https://storage.googleapis.com/tensorflow/keras-applications/vgg16/vgg16_weights_tf_dim_ordering_tf_kernels_notop.h5

58889256/58889256 ————— **0s** 0us/step

Model: "sequential"

Layer (type)	Output Shape	Param #
rescaling (Rescaling)	?	0 (unbuilt)
vgg16 (Functional)	(None, 7, 7, 512)	14,714,688
global_average_pooling2d (GlobalAveragePooling2D)	?	0
dense (Dense)	?	0 (unbuilt)
dense_1 (Dense)	?	0 (unbuilt)

Total params: 14,714,688 (56.13 MB)

Trainable params: 0 (0.00 B)

Non-trainable params: 14,714,688 (56.13 MB)

Epoch 1/5

4/4 ————— 28s 7s/step - accuracy: 0.5938 - loss: 0.6942 -
val_accuracy: 1.0000 - val_loss: 0.5708

Epoch 2/5

4/4  **29s** 8s/step - accuracy: 1.0000 - loss: 0.5158 -

val_accuracy: 1.0000 - val_loss: 0.4540

Epoch 3/5

4/4  **37s** 6s/step - accuracy: 1.0000 - loss: 0.4224 -

val_accuracy: 1.0000 - val_loss: 0.3593

Epoch 4/5

4/4  **24s** 6s/step - accuracy: 1.0000 - loss: 0.3213 -

val_accuracy: 1.0000 - val_loss: 0.2669

Epoch 5/5

4/4  **46s** 8s/step - accuracy: 1.0000 - loss: 0.2367 -

val_accuracy: 1.0000 - val_loss: 0.2064

1/1  **4s** 4s/step - accuracy: 1.0000 - loss: 0.2064

Accuracy: 1.0